

On the non-smoothness of stress components in the dynamic torsion verification problem

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Summary

For the dynamic torsion benchmark, the individual Cauchy stress *components* ($\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$, and to a lesser extent the off-diagonal terms) appear “choppy” in space, while the von Mises stress is smooth. This is the *physically correct* behavior, not a numerical artifact, a projection error, or a deficiency of Norma. Two facts explain it:

1. **It is a strongly dynamic, wave-dominated problem.** The bar is set in motion by a large initial torsional velocity (corner speed ≈ 141 m/s). Elastic waves propagate and reflect off the free ends throughout the simulation. The small normal stress components are dominated by these traveling waves and therefore oscillate in space.
2. **The response is shear-dominated.** The twisting generates shear stresses ≈ 87 MPa, an order of magnitude larger than the normal components (~ 6 MPa). Von Mises is governed by the (smooth, large) shear and is essentially blind to the (small, oscillatory) normal components.

We confirm this with a four-code verification: **Norma**, **Carina**, **LCM/Albany**, and **Sierra/Adagio** all reproduce the same behavior on the same mesh, with peak von Mises agreeing to within 3% (Norma and Carina agree to machine precision). The choppiness is present in *every* code; it is physics, not a bug.

1 The problem is highly dynamic

A $0.05 \times 0.05 \times 1.0$ m Neo-Hookean bar ($E = 1$ GPa, $\nu = 0.25$, $\rho = 1000$ kg/m³), free at both ends, is given the initial torsional velocity field $v_x = -a y z$, $v_y = a x z$, $v_z = 0$ with $a = 8000$ (m s)⁻¹ (so that a multiplied by two lengths yields a velocity). It is integrated implicitly (Newmark, $\beta = 0.25$, $\gamma = 0.5$, $\Delta t = 1$ μ s) to $t = 2$ ms.

The initial velocity is large: at a top corner ($z = 0.5$), $|\mathbf{v}| = \sqrt{2} a (0.025)(0.5) \approx 141$ m/s. The bar twists through roughly half a turn end-to-end at maximum deformation. The relevant wave speeds are

$$c_s = \sqrt{G/\rho} = 632 \text{ m/s}, \quad c_p = \sqrt{(K + \frac{4}{3}G)/\rho} = 1095 \text{ m/s}, \quad (1)$$

with $G = 4.0 \times 10^8$ Pa, $K = 6.67 \times 10^8$ Pa. Over the bar length $L = 1$ m these give transit times $L/c_s = 1.58$ ms and $L/c_p = 0.91$ ms, so in the 2 ms window the shear and dilatational fronts cross the bar ~ 1.3 and ~ 2.2 times, respectively, reflecting repeatedly off the free ends. *The bar is full of propagating, reflecting elastic waves at all times.*

2 Why the stress components are non-smooth

In pure (small-strain) torsion the only nonzero stresses are shears; the normal components vanish. Here the normal components $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$ are nonzero only through (i) finite-deformation coupling (second order in the twist) and (ii) the axial/dilatational waves launched by the impulsive start and the free-end reflections. They are therefore both *small* ($\sim 1\text{--}6\text{ MPa}$) and *dominated by traveling waves*. A wave field sampled at a single instant is spatially oscillatory by nature — the “choppiness” is the snapshot of wavefronts moving through the bar. Because the normal components have no large, smooth mean value to sit on, the wave ripple *is* the entire signal, so it is fully visible.

3 Why von Mises is smooth

The von Mises stress is

$$\sigma_{\text{vM}} = \sqrt{\frac{1}{2}[(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2] + 3(\sigma_{xy}^2 + \sigma_{yz}^2 + \sigma_{xz}^2)}. \quad (2)$$

At peak deformation $\sigma_{\text{vM}} \approx 150\text{ MPa}$, which for this shear-dominated state corresponds to a shear magnitude $\tau \approx \sigma_{\text{vM}}/\sqrt{3} \approx 87\text{ MPa}$. The shear term $3\tau^2$ thus exceeds the normal-difference term by roughly three orders of magnitude ($3 \cdot 87^2$ vs. a few $\times 6^2$, in MPa^2). Since σ_{vM} scales with the *square* of the components, the small oscillatory normal stresses perturb it by well under 1%. Von Mises therefore tracks the large, smooth, twist-driven shear and looks smooth — exactly as observed. This is a property of the quantity, not evidence that the components are “more accurate” than von Mises or vice versa.

4 Is it the coarse mesh?

The mesh ($2 \times 2 \times 40$ hexes) is coarse, and that does set the *granularity* of what is plotted: the raw quadrature-point values are piecewise discontinuous between elements (hence the blockiest pictures), and a coarse mesh resolves only the longer-wavelength waves, aliasing the shorter ones. The L^2 projection to nodes removes the inter-element jumps but cannot manufacture sub-element information, so the projected normal components still show the underlying oscillation. Crucially, this is *resolution*, not *error*: refining the mesh would resolve the waves more finely (more, and crisper, wavefronts) and converge to the true wave-filled solution — it would not flatten the components to zero, because the oscillation is real. The clinching evidence is in the next section: four independent codes on this *same* coarse mesh produce the same fields.

5 Four-code verification

The identical problem was run in four independent codes on the same mesh, with stresses projected to the nodes by L^2 projection and evaluated at the instant of maximum strain energy ($t^* = 7.9 \times 10^{-4}\text{ s}$).

Figure 1 overlays the projected nodal fields along the bar for all four codes. Two things are evident. First, von Mises is a smooth tent peaking at mid-span, while the normal components are oscillatory. Second, the curves for **Norma**, **Carina**, and **LCM** lie **essentially on top of one another**: the oscillation is reproduced, to plotting accuracy, by three completely independent implementations, so it is the physical solution and not code-specific noise.

Adagio shows the same qualitative pattern but is visibly offset from the other three. This is expected, and it is *not* a discrepancy in the stress computation: Adagio uses a *lumped* mass

Code	Framework	Peak σ_{vM} [MPa]	Agreement vs. Norma
Norma	Julia FE (reference)	150.0	—
Carina	Julia FE (GPU)	150.0	machine precision ($\sim 10^{-4}$ Pa)
LCM	C++/Trilinos (Albany)	150.1	0.09%
Adagio	Sierra/SolidMechanics	155.1	$\sim 3\%$ (lumped mass, see below)

Table 1: Peak nodal von Mises at t^* and node-wise agreement with Norma (matched by reference coordinates). Norma and Carina (sibling codes, consistent mass) agree to machine precision and LCM to 0.09%. Adagio’s $\sim 3\%$ difference comes from its *lumped* mass matrix for implicit integration: the lumped mass changes the rotational inertia and hence the dynamic trajectory, so at the fixed time t^* Adagio is in a slightly different configuration. Its stress projection is consistent — the difference is in the time-integration mass, not the stress.

matrix for the implicit integration, whereas the other three use the *consistent* mass. The lumped mass changes the rotational inertia of the discretization and therefore the dynamic trajectory of the twisting bar. By the fixed sampling time t^* the Adagio bar is consequently in a slightly different twist configuration than the consistent-mass codes, so its instantaneous stresses are sampled at a correspondingly different state — which is exactly why its curves are shifted rather than coincident. The three consistent-mass codes share the same trajectory and so coincide. (Adagio’s stress projection itself is consistent; the difference is purely in the time-integration mass.)

Figure 2 shows the smooth von Mises field on the deformed (twisted) bar, and Figures 3–5 the normal components, which display the propagating-wave pattern. The same coincidence holds: the field is visually identical for the three consistent-mass codes, with Adagio sampled at its slightly different configuration.

6 Conclusion

The non-smoothness of the stress components is the correct physical response of a coarse-mesh, strongly dynamic, wave-filled torsion problem; von Mises looks smooth because the response is shear-dominated and von Mises is governed by that large, smooth shear. Four independent codes reproduce both features on the same mesh and agree on peak von Mises to within 3% (two of them to machine precision). None of this indicates any inaccuracy in Norma.

Four-way torsion verification — nodal L2-projected stress at max strain energy ($t^* = 7.900\text{e-}04$ s)

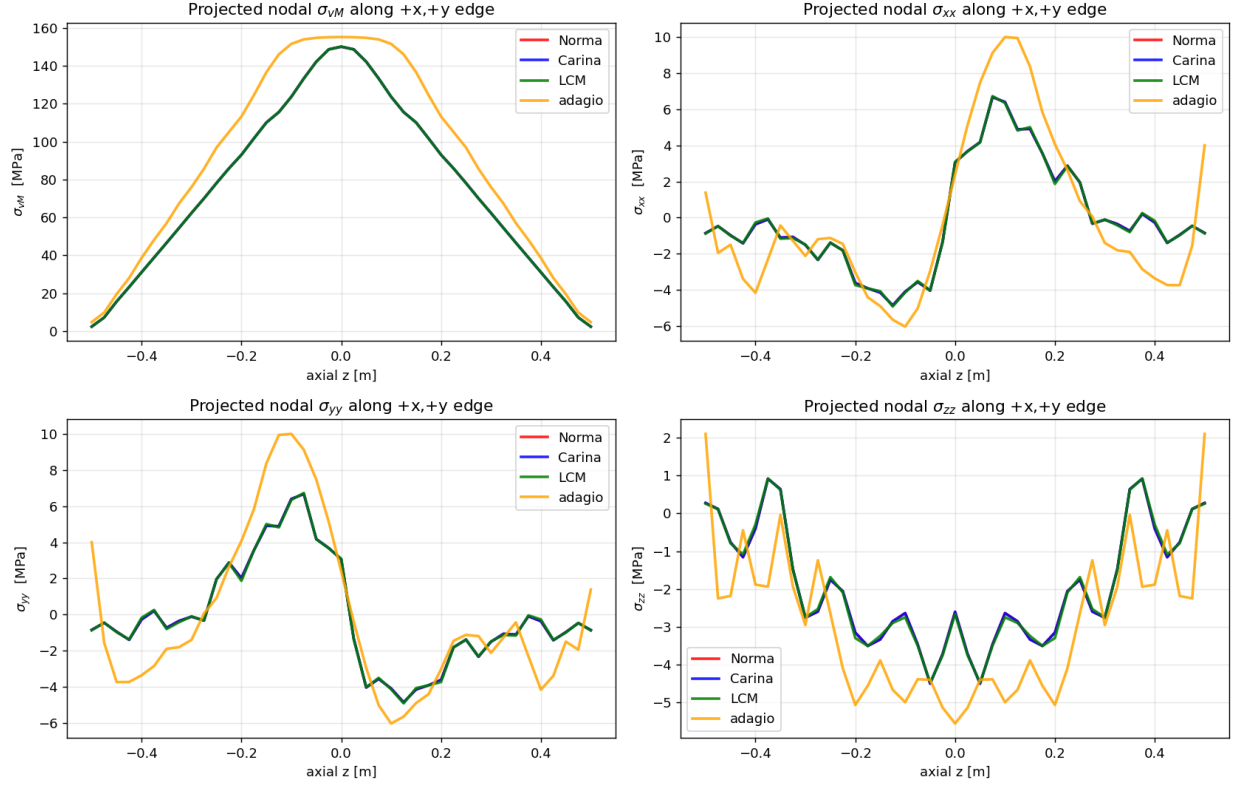


Figure 1: Projected nodal stress vs. axial position along the bar's edge, four codes overlaid (Norma red, Carina blue, LCM green, Adagio orange). Von Mises (top left) is smooth; the normal components are wave-dominated and oscillatory, yet the four codes agree.

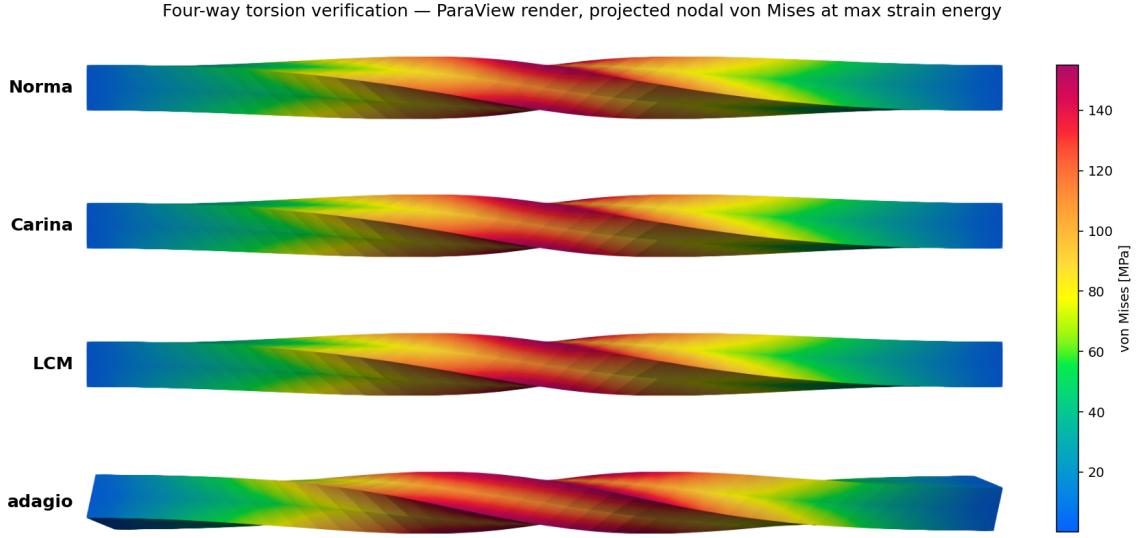


Figure 2: ParaView render of the twisted bar at t^* (true scale), projected nodal von Mises, four codes stacked. The field is smooth in every code; Norma, Carina, and LCM are visually identical, while Adagio (lumped mass) is sampled at a slightly different configuration.

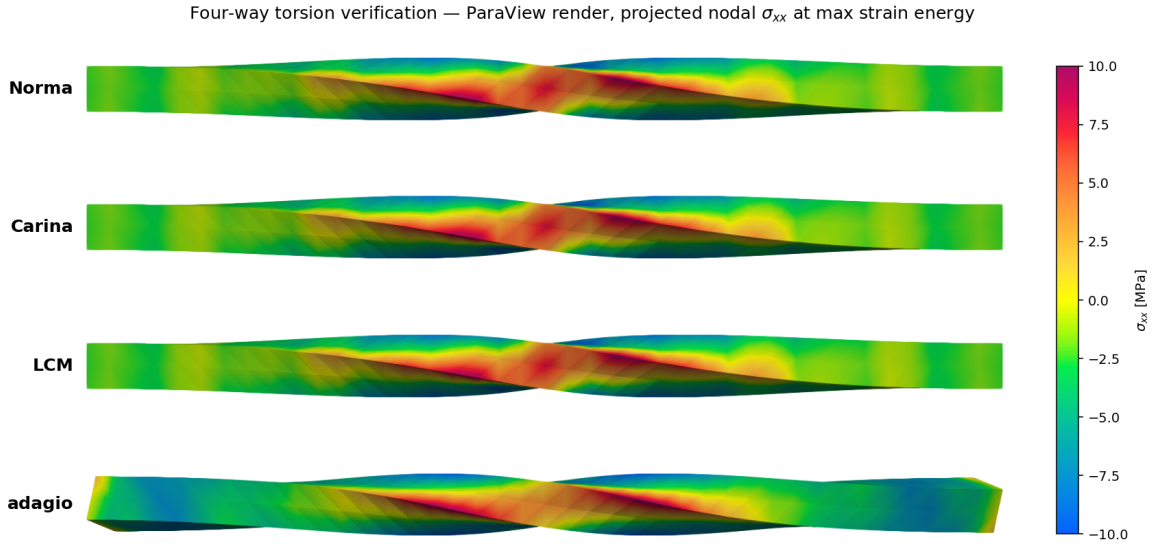


Figure 3: Projected nodal σ_{xx} on the twisted bar at t^* . The propagating-wave pattern is evident. Norma, Carina, and LCM agree; Adagio (lumped mass) is sampled at a slightly different configuration.

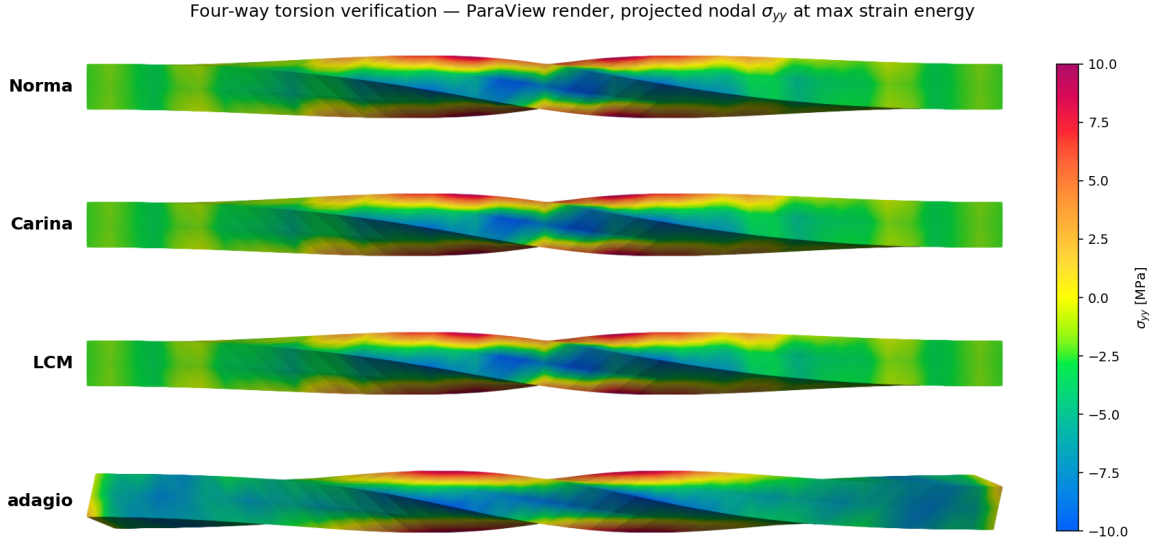


Figure 4: Projected nodal σ_{yy} on the twisted bar at t^* , four codes stacked. As for σ_{xx} , the wave pattern is physical and shared by the three consistent-mass codes.

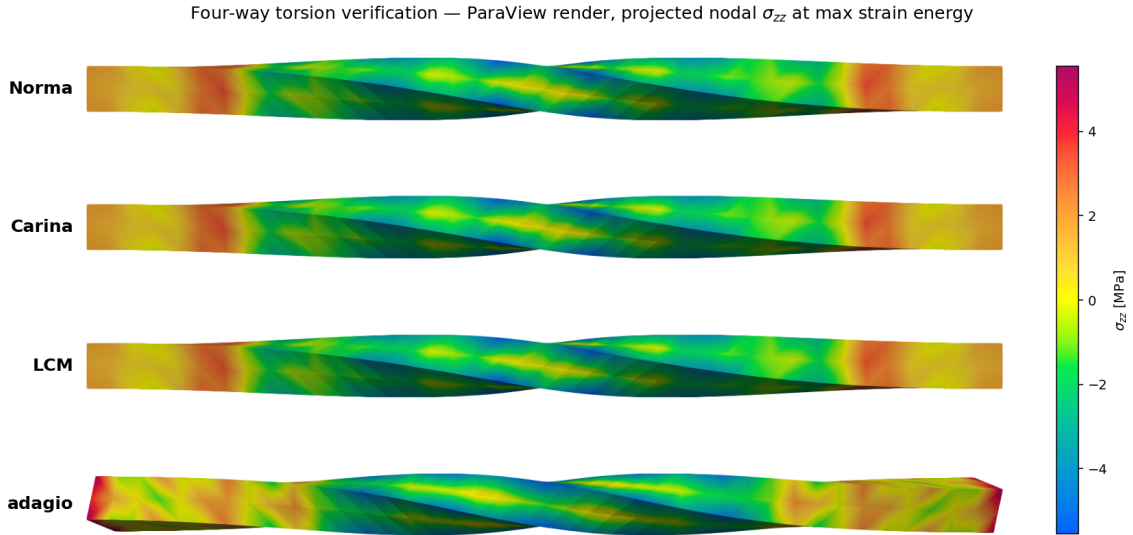


Figure 5: Projected nodal σ_{zz} on the twisted bar at t^* , four codes stacked. The axial normal stress likewise reflects the dilatational waves propagating along the bar.